POLITECNICO MILANO 1863 | DIPARTIMENTO DI MECCANICA

Surrogate-Assisted Compliant Base-Arm Optimization

Surrogate Modelling · Bayesian Optimization · True-EoM Validation

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Now we starts the presentation about our Python project on surrogate-assisted optimization of a compliant base-arm system.

Contents

1. **Problem & Dynamic Model**

   Target quantity · Chirp excitation

2. **Neural Network Surrogate**

   Dataset · Outputs · Performance

3. **Neural Network vs Physics-Informed Neural Network**

   Comparison · Final surrogate choice

4. **Bayesian Optimization**

   Cost function · Nominal BO result · True-EoM validation

5. **Robustness Validation**

   Base · Base-specific BO · Arm · Initial conditions

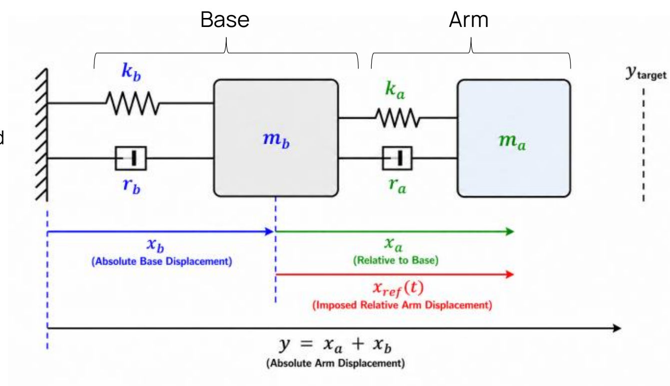
6. **Conclusions**

The presentation contents follow the project development workflow, as presented in the slide.

Problem & Dynamic Model

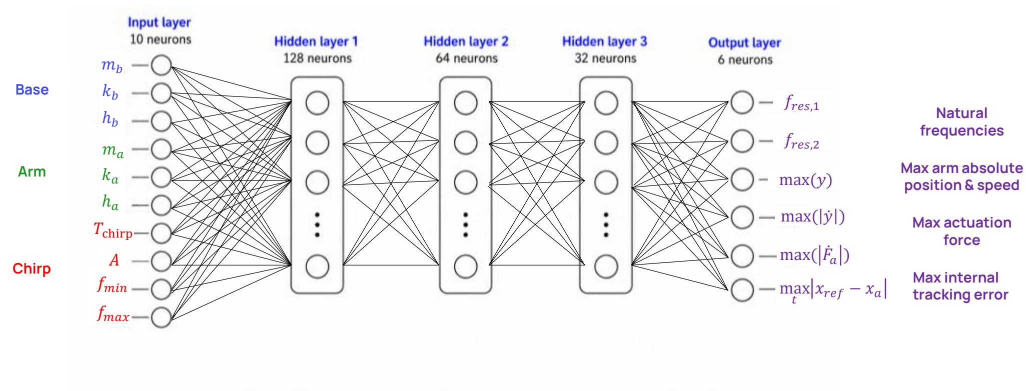
System Objective
- Compliant base–arm system
- Controlled quantity: $y = x_a + x_b$
- Target requires dynamic amplification: $y_{target} > A_{max}$
- Chirp reference $x_{ref}(t)$ excites the coupled dynamics

Design Challenge
- Each candidate requires time-domain EoM simulation
- Constraints on target band, velocity and actuation force
- NN surrogate used to enable efficient BO search



The system is a two-degree-of-freedom compliant base-arm model.
The controlled quantity is the absolute arm displacement, y as the sum of the arm displacement relative to the base and the absolute base displacement.
The target is intentionally larger than the maximum allowed command amplitude, so the solution cannot be purely quasi-static.
The command must exploit dynamic amplification, and this is why the input is chosen as a chirp, able to excite a frequency interval of the coupled system.

Neural Network Surrogate - Structure



Fully connected MLP: 10 → 128 → 64 → 32 → 6 | **ReLU hidden layers**

The neural network is introduced to avoid integrating the equations of motion for every candidate during optimization.
Its input are the base, arm, and chirp parameters.
Its outputs are the quantities needed later by BO, so the two natural frequencies, peak displacement, peak velocity, peak actuation force, and maximum internal tracking error.
Since these are scalar KPIs rather than time histories, a compact fully connected Multi Layer Perceptron is sufficient.

Neural Network Surrogate - Generation of Training Dataset

Sampling Strategy
- Latin hypercube sampling
- Full-domain sampling to avoid first-mode bias

Dataset Size
- 10 000 accepted EoM simulations

Chirp Validity Condition

$$f_{max} > f_{min} + 0.05\ Hz$$

Input	Meaning	Sampling range	Unit
m_b	Base mass	$[56, 84]$	kg
k_b	Base stiffness	$[3200, 4800]$	N/m
h_b	Base damping ratio	$[0.20, 0.30]$	–
m_a	Arm mass	$[8, 12]$	kg
k_a	Arm stiffness	$[100, 5000]$	N/m
h_a	Arm damping ratio	$[0.20, 1.10]$	–
f_{min}	Chirp lower frequency	$[0.05, 2.00]$	Hz
f_{max}	Chirp upper frequency	$[0.20, 10.00]$	Hz
T_{chirp}	Chirp duration	$[5, 15]$	s
A	Chirp amplitude	$[0.01, 0.10]$	m

The surrogate is trained over the full admissible input domain, so the BO search is not pre-biased toward a prescribed resonance region.

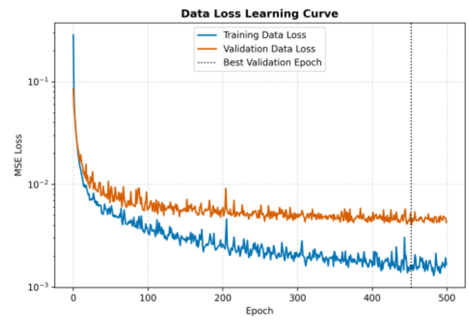
The dataset is generated with Latin Hypercube Sampling over the full admissible domain.
This is important because the surrogate should not be pre-biased toward a prescribed resonance region before BO starts.
Each accepted point is evaluated with the true equations of motion, and the final dataset contains 10,000 valid simulations.
The only imposed validity condition is that the upper frequency must be meaningfully larger than the lower one.

Neural Network Surrogate – Training & Test Accuracy

Training Setup
- Dataset: 10,000 accepted samples
- Train/Validation/Test split: 70/15/15
- Adam optimizer, MSE loss
- Learning rate: 2×10^{-3}
- Best validation checkpoint used for testing

Output	R^2	MAPE [%]
$f_{res,1}$	0.9979	0.34
$f_{res,2}$	0.9993	0.67
$\max(y)$	0.9851	3.40
$\max(\lvert\dot{y}\rvert)$	0.9979	3.59
$\max(\lvert F_a\rvert)$	0.9951	9.18
$\max(\lvert x_{ref} - x_a\rvert)$	0.9974	3.68

Most sensitive KPI: actuation-force peak.



No severe overfitting: validation loss remains close to training loss.

! The NN is accurate enough to guide BO, but final feasibility is checked with **true-EoM validation**.

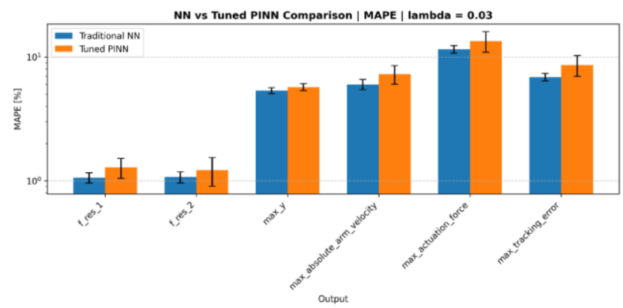
The final network is trained with a 70/15/15 split and the best validation checkpoint is used for testing.
The learning curves show that the validation loss remains close to the training loss, so there is no evidence of severe overfitting.
Overall, the test metrics are accurate enough for the intended role of the surrogate, so guiding the BO search toward promising regions of the design space.
The most sensitive output is the force peak, which is why force feasibility is never accepted from the NN alone and is always checked again with the true EoM.

NN vs Physics-Informed NN

Physics-informed test

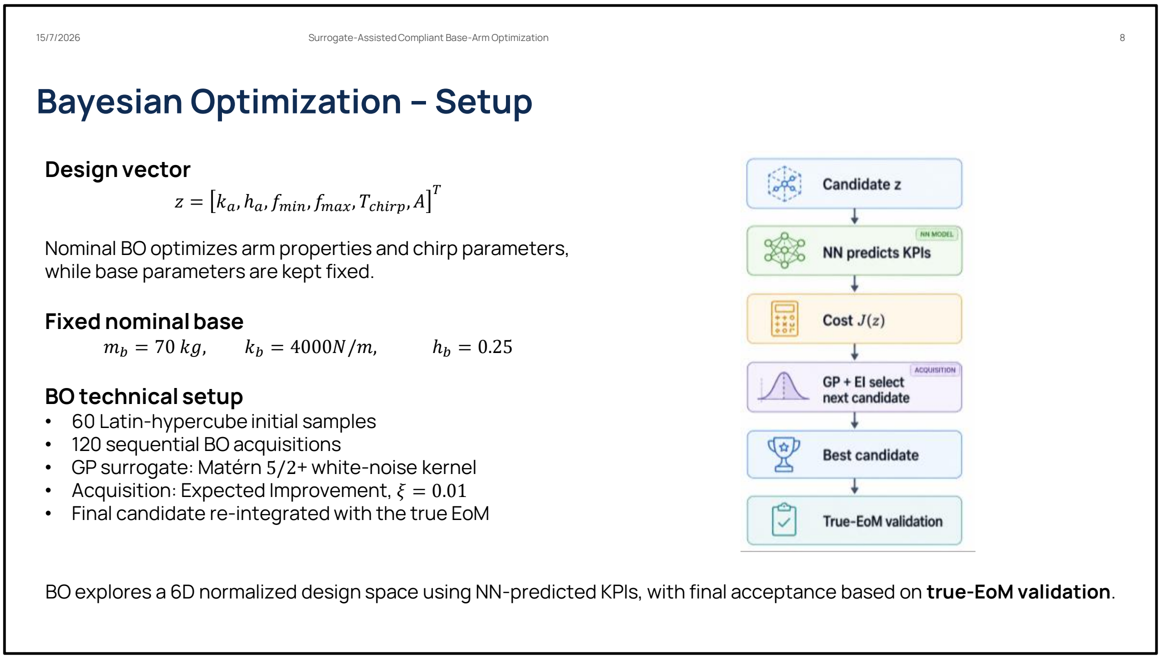
$$\mathcal{L}_{tot} = \mathcal{L}_{data} + \lambda_{phys}\mathcal{L}_{phys}$$

- Physics residual applied only to natural frequencies
- λ_{phys} sweep in $[0, 0.5]$, repeated over 3 seeds
- Same architecture, dataset split and training setup
- Final NN vs tuned PINN comparison over 5 seeds: $\lambda_{phys} = 0.03$

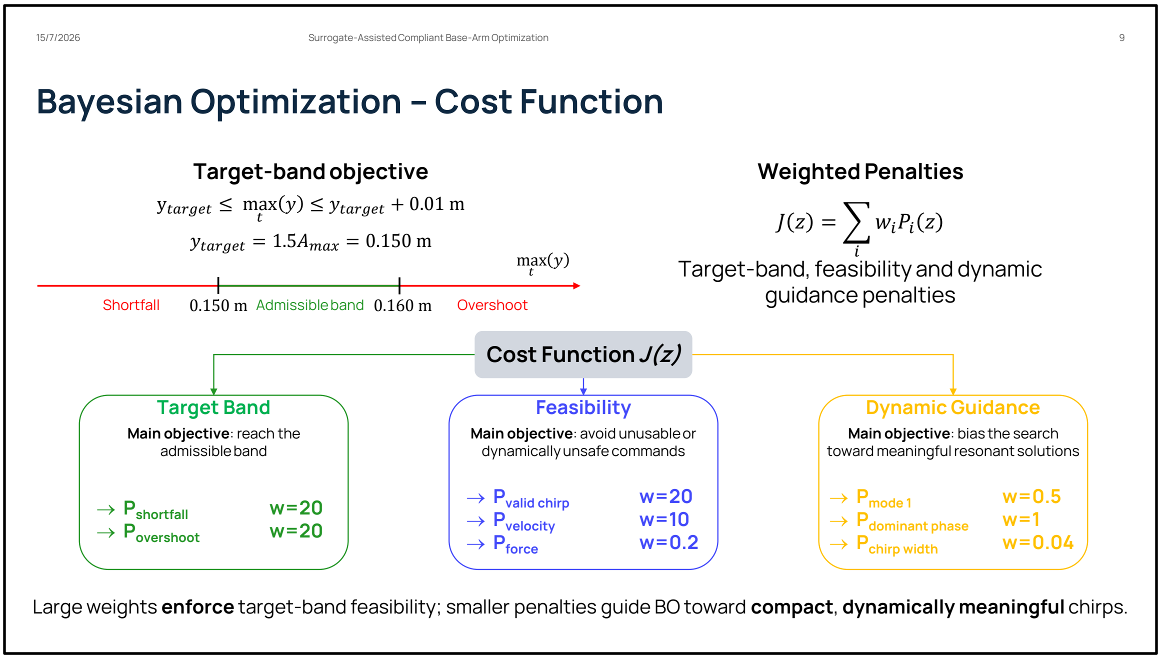


Traditional NN retained for BO: the physics residual **did not improve** the dynamic KPIs.

A physics-informed variant was tested by adding an eigenfrequency residual on the two natural frequency outputs.
This was a reasonable idea, but it constrained only part of the output vector, while the optimizer also depends on dynamic KPIs such as displacement, velocity, force, and tracking error.
In the controlled comparison, the traditional NN performed slightly better on all outputs, therefore, for this pipeline, we retained the traditional NN as the BO surrogate.



- After selecting the traditional **neural network**, we use it as a **black box fast evaluator inside** the **Bayesian Optimization** loop. **The outputs are used to compute the cost function**
- **Design vector**
  - Arm mass is fixed
  - Arm damping, stiffness optimized together with chirp lower and upper frequency, duration and amplitude
- At this stage, the compliant **base** is kept **fixed** at its nominal configuration.



- **Cost function: weighted sum of different contributions divided in 3 groups**
- **Target band**
  - Guarantee that the absolute arm displacement falls within an admissible band
- **Feasibility**

  - P_valid_chirp prevents reversed or nearly degenerate frequency intervals.
  - P_velocity discourages excessively aggressive dynamic responses
  - P_force discourages high actuation loads.
- **Dynamic Guidance**
  - P_mode 1 drives the chirp towards the first mode
  - P_dominant phase avoids an unfavourable base–arm contribution where the amplification is strongest.
  - P_chirp width acts as a weak tie-breaker against unnecessarily broad frequency sweeps.
- The penalties were first **normalized,** the weights were assigned according to a **hierarchy** of requirements and **tuned** with **repeated BO** runs

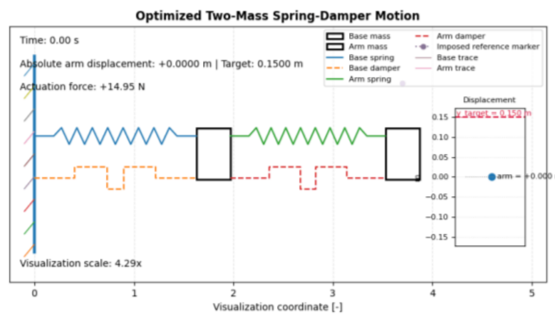
Giustificazione limite forze e rispettivi pesi

- The 150-newton **force limit** is a literature-informed **scale based** on the **reference paper**.
- The 2-metre-per-second **velocity limit** is a **project-level safety** and regularity bound.
- Velocity receives a higher weight because it directly measures the dynamic aggressiveness of the chirp response.
- The **force penalty** is **softer** because force was **not the active limitation** during tuning. Larger force weights produced **more conservative solutions** and could reduce target reaching.

Bayesian Optimization – Nominal Candidate

Optimized parameters

Candidate	k_a [N/m]	h_a [-]	f_{min} [Hz]	f_{max} [Hz]	T_{chirp} [s]	A [m]
Target-band BO	289.02	0.5085	0.436	1.347	12.664	0.09976



Validated performance

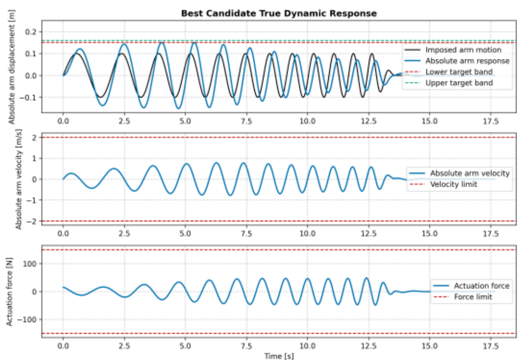
- Target response: $\max_t(y) = 0.15060$ m
- Velocity: $v_{max} = 0.778$ m/s < 2 m/s
- Force: $F_{max} = 48.43$ N < 150 N

The nominal BO candidate **reaches** the target band while keeping velocity and force **well below their limits**.

- **Optimized design variables**
- **Outputs of the system computed by re-integrating the EOMs**

- **Validated performance**
  - The values shown here are **not only neural-network predictions:** they are obtained after **re-integrating** the selected candidate with the **true equations of motion**.
- The imposed arm motion has an amplitude of approximately 0.10 metres, while the absolute arm response reaches 0.1506 metres → **target is reached through the dynamic amplification of the** coupled base–arm **system**, rather than by directly imposing a command equal to the target displacement
- Maximum velocity and maximum force well below the imposed limits

Bayesian Optimization – True-EoM Validation



Time-domain evidence

- Peak displacement inside the target band
- Velocity far below the $\pm\mathbf{2\ m/s}$ limit
- Force far below the $\pm\mathbf{150\ N}$ limit
- Fade-out avoids abrupt command termination

Validation margins

- Shortfall margin: $+0.60\ \mathrm{mm}$
- Overshoot margin: $+9.40\ \mathrm{mm}$
- Velocity margin: $1.22\ \mathrm{m/s}$
- Force margin: $101.57\ \mathrm{N}$

The time histories confirm target-band reaching **without** hidden transient **violations**.

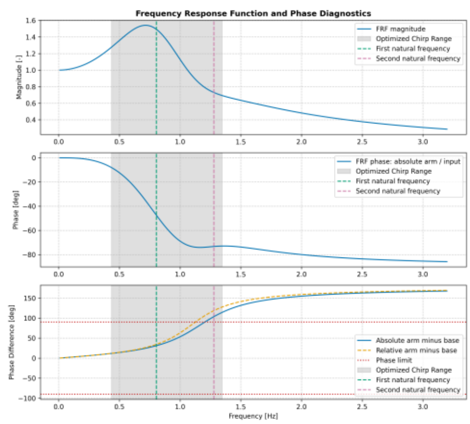
- **Time histories of displacement, velocity, force**
- **Validation margins**
  - **How far is the solution from violating the constraints**
  - **Shortfall margin very small compared to overshoot margin**
  - High margins on speed and velocity
- Short smooth fade-out is then appended to avoid an abrupt interruption of the reference command
- The fade-out is not part of the optimized design vector and does not artificially produce the target peak.

Bayesian Optimization – Frequency Domain Interpretation

Key diagnostics
- Optimized chirp range: $[0.436, 1.347]\ Hz$
- Dominant frequency: $f_{dom} = 0.715\ Hz$
- Dominant phase check: $|\ \Delta\phi_{y,b}(f_{dom})\ | = 24.65° < 90°$
- First-peak dominance: $G_2/G_1 = 0.876 < 1$

FRF-time domain consistency
- FRF gain: $G(f_{dom}) = 1.541$
- Effective amplification: $\max_t(y)\ /A_{chirp} = 1.510$

! The optimized chirp excites the dominant amplification region while keeping the base–arm phase within the adopted **90°threshold**.



- **Explains why is the chirp effective**
- **Dominant frequency**: where **FRF** reaches its **maximum**
- First peak/mode is dominant but second one is not negligible → **chirp still includes both natural frequencies**
- At the dominant frequency, the absolute arm-minus-base phase difference is well below the limit value
- **The maximum FRF gain inside the chirp interval is 1.541. The effective amplification measured in the time-domain simulation is 1.510. → chirp sweep is sufficiently slow to guarantee quasi steady state response**

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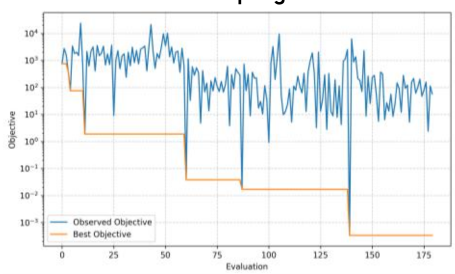
Bayesian Optimization – Search Diagnostics

BO search progress



Best surrogate objective progressively decreases during BO acquisition.

Final-candidate surrogate error

Quantity	Relative error [%]
$f_{res,1}$ [Hz]	2.34
$f_{res,2}$ [Hz]	1.76
$\max(y)$ [m]	2.16
v_{max} [m/s]	6.41
F_{max} [N]	47.59
Maximum tracking error [m]	6.20

NN guides the search; true EoM decides final feasibility. **Force-peak prediction** remains the most **critical** surrogate output.

Final candidate surrogate error

- The table on the right compares the **neural-network predictions** with the **true-EoM results** for the final candidate.
- The NN predicts the final candidate well enough to guide the search but cannot be used in the validation phase
- In this candidate the **force is overestimated** (**verified** by looking at the values), which is conservative, but this **cannot be assumed everywhere → true-EoM validation remains decisive in the final feasibility analysis.**

Robustness Validation – Base Stress-Test Setup

Purpose

- Test the fixed nominal BO command under base-parameter variations
- Keep optimized arm and chirp unchanged
- Identify where the nominal command remains admissible

Stress test, not a statistical uncertainty model

Stress-test grid

- $m_b/m_{b,0} \in [0.25, 1.75]$ 13 levels
- $k_b/k_{b,0} \in [0.25, 1.75]$ 13 levels
- $h_b/h_{b,0} \in \{0.25, 0.625, 1, 1.375, 1.75\}$

13×13×5=845

true-EoM simulations

Margin-based admissibility rule

$$\mathbf{0 \le M_y, \le 0.01}\ \mathrm{m}, \qquad \boldsymbol{M_v, M_F, M_\phi \ge 0}$$

Target inside band · Velocity below limit · Force below limit · Phase below limit

The campaign maps the validity domain of the fixed nominal command under base-dynamics variations.

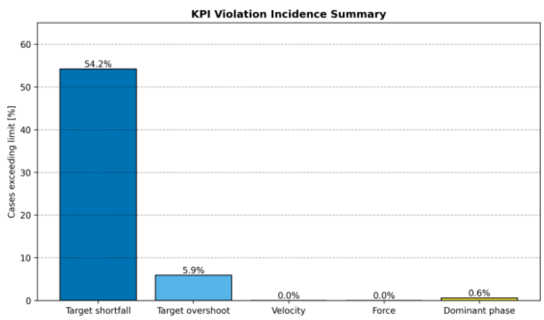
Purpose:

- **Arm + chirp** parameters **fixed**
- **Base** parameters **perturbed**
- The grid maps **failure boundaries**; it does **not estimate failure probability**

Grid:

- **m_b0** base parameters used for the **nominal BO**
- **Full factorial** design
- The **neural-network** surrogate is **not used** to decide admissibility, because this stage is focused on **accurate physical validation** rather than fast design-space exploration
- Only 5 damping levels to be able to represent the results in clear way (only 5 windows)

Base Perturbation – Dominant Failure Modes

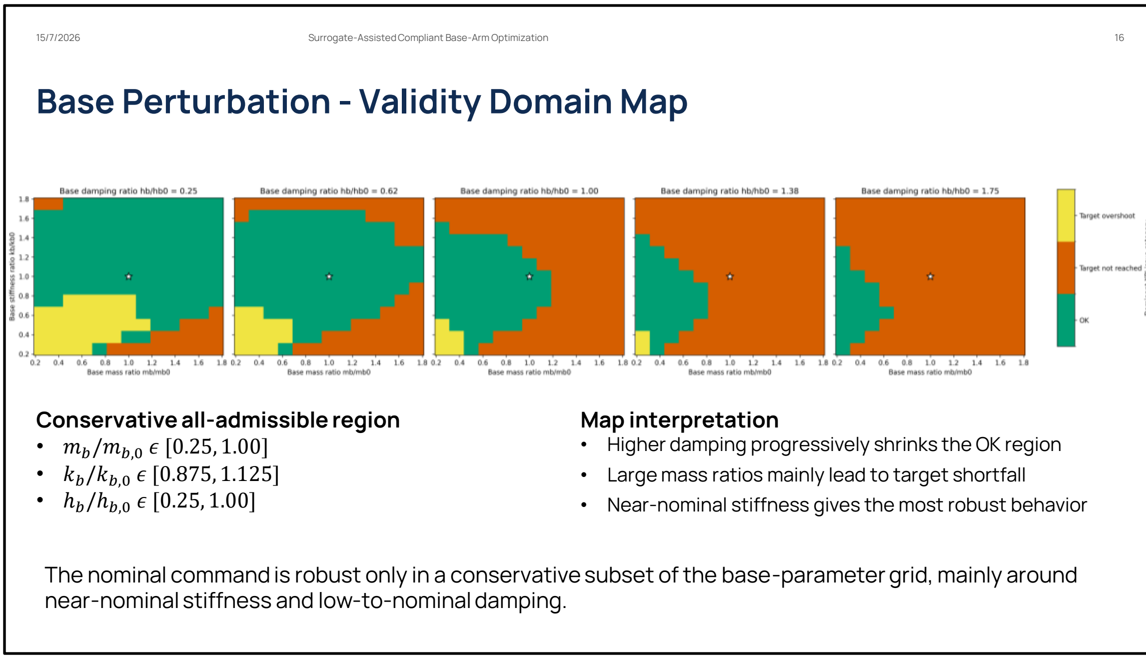


Failure interpretation

- Target shortfall is the dominant issue
- Target overshoot occurs only in a limited portion of the grid
- Velocity and force limits are never activated
- Phase violation is negligible

Nominal-command robustness is mainly limited by **insufficient target reaching**, not by velocity or force saturation.

- The **dominant failure** mode is **target shortfall (consistent** with the small **reaching margin** of the nominal solution)
    - **base-parameter variations reduce** the **dynamic amplification** exploited by the nominal command
- **Different violations** may occur in the **same configurations** → the percentages in this plot should not be added directly
    - Example: all five phase-critical cases are also included among the target-shortfall cases.

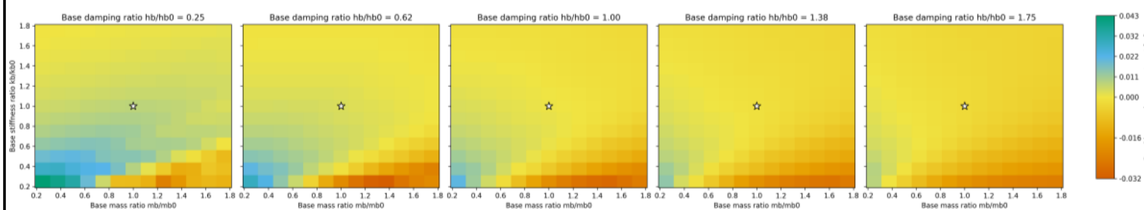


- The validity maps show where the fixed command remains admissible. Each panel corresponds to one damping level, while the axes are base mass ratio and base stiffness ratio
- Increasing **damping** makes the **admissible region smaller**, because more **energy** is **dissipated**
- Large mass ratios mainly lead to shortfall.
- **Stiffness** has a more **coupled** effect,

FRF peaks

- Changing the base parameters also shifts and reshapes the FRF peaks.
- Therefore, **peak containment** is **necessary but not sufficient** to guarantee that the target band and all other **requirements** remain **satisfied**.
- In all the failed cases the natural frequencies were included within the chirp range

Base Perturbation - Reaching Margin



Margin definition

$$M_y = \max_t(y) - y_{target}$$

- $M_y < 0$ $\longrightarrow$ target shortfall
- $0 \leq M_y \leq 0.01$ m $\longrightarrow$ admissible target band
- $M_y > 0.01$ m $\longrightarrow$ target overshoot

Sensitivity trend

Spearman correlation

- $\rho(M_y, m_b/m_{b,0})$ = -0.51
- $\rho(M_y, h_b/h_{b,0})$ = -0.57
- $\rho(M_y, k_b/k_{b,0})$ = +0.04

Higher base mass and damping reduce the reaching margin; base stiffness acts through coupled dynamics rather than a simple monotonic trend.

- **<u>Heatmap quantifies how much does the reaching margin change</u>**
- **Sensitivity trend:**
  - Spearman correlation coefficient → monotonicity
  - Raching margin decreases with increasing mass
  - Reaching margin decreases with increasing damping
  - No strong monotonic trend with stiffness

Base Perturbation – Base-Specific Re-Optimization

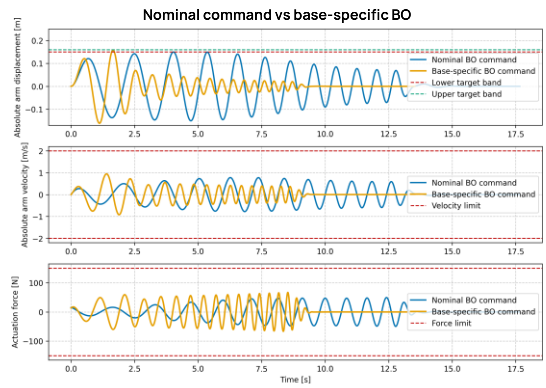
Selected perturbed base

- $m_b/m_{b,0} =$ 1.125
- $k_b/k_{b,0} =$ 1.125
- $h_b/h_{b,0} =$ 1.00
- **Nominal-command failure**: target shortfall

Candidate	M_y [mm]	M_v [m/s]	M_F [N]	Admissible
Nominal BO	-0.174	1.221	100.42	NO
Base-Specific BO	7.711	1.055	82.99	YES

What changed after re-optimization?

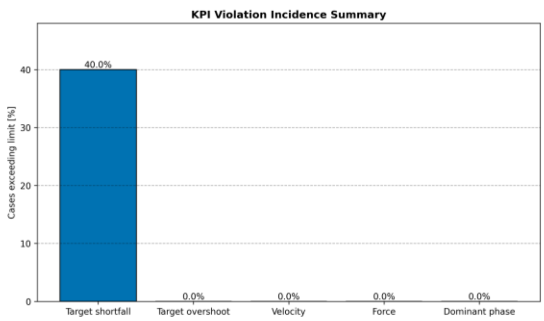
- Chirp range shifted to higher frequencies: $\mathbf{0.436–1.347\ Hz \rightarrow 0.699–2.826\ Hz}$
- Chirp duration shortened: $\mathbf{12.66\ s \rightarrow 8.53\ s}$
- Amplitude almost unchanged: $\mathbf{0.09976\ m \rightarrow 0.09948\ m}$



The perturbed base remains feasible: base-specific BO restores target reaching through a different chirp strategy, not a larger input amplitude.

- <u>This slide checks **whether a base configuration that fails with the nominal command is actually infeasible, or whether it only requires re-optimization.**</u>
- <u>With the nominal BO command, this perturbed base gives a small target shortfall.</u>
- When BO is run again for that same base, the response becomes admissible
- The **recovery** is **not obtained by** simply **increasing** the **amplitude**, which remains almost unchanged. Instead, the chirp **range** and **duration change**, showing that feasibility can be recovered through a **different dynamic excitation strategy**.

Arm Perturbation – Dominant Failure Modes



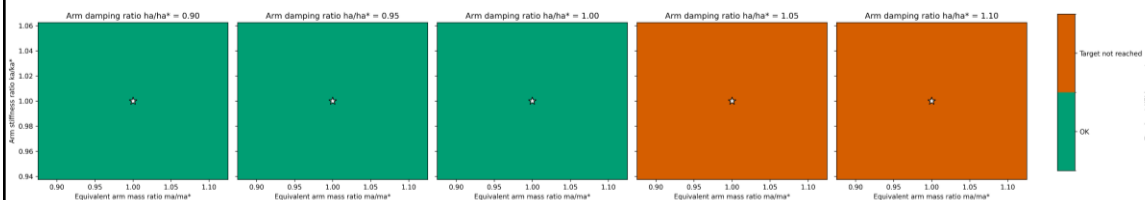
Local validation setup
- Fixed nominal base and optimized chirp
- Perturbed arm mass, stiffness and damping
- $5 \times 5 \times 5 = 125$ true-EoM simulations
- Local validation, not a broad stress test

Failure interpretation
- Target shortfall is the only active failure mode
- 75/125 cases remain admissible
- No overshoot, velocity, force or phase violation
- Robustness loss is driven by reduced amplification

Arm-parameter robustness is limited only by insufficient target reaching, mainly caused by increased arm damping.

The second test campaign was performed on the arm, changing its parameters locally and keeping the base and the chirp fixed.
This is a local validation consisting of 125 simulations, with 75 successful.
As we can see, every single failure is a "target-shortfall", so the only problem is insufficient amplification

Arm Perturbation – Validity Domain Map



Verified all-admissible local region

- $m_a/m_a^* \epsilon [0.90, 1.10]$
- $k_a/k_a^* \epsilon [0.95, 1.05]$
- $h_a/h_a^* \epsilon [0.90, 1.00]$

Discrete verified region from the tested grid.

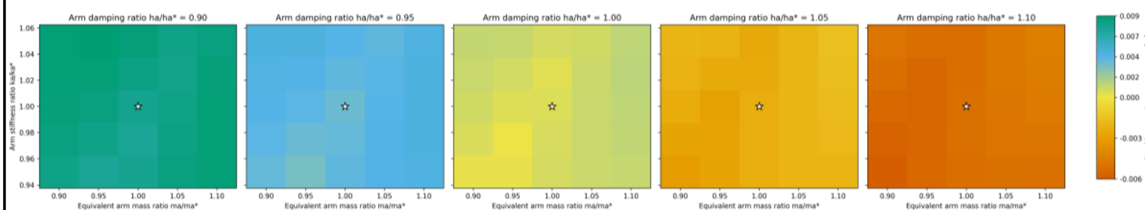
Map interpretation

- Validity is mainly damping-limited
- Mass and stiffness remain locally tolerated
- Increasing h_a reduces dynamic amplification
- The boundary lies between $h_a/h_a^* = 1.00$ and 1.05

The nominal command is locally robust to arm mass and stiffness perturbations, But loses admissibility when arm damping increases above the nominal value.

This slide displays a validity map that immediately reveals a clear trend.
While changes in mass and stiffness are generally well-tolerated,
increasing the arm damping above nominal values leads to system failure.

Arm Perturbation – Reaching Margin



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Sensitivity trend

Spearman correlation

- $\rho(M_y, m_a/m_a^*)$ = +0.10
- $\rho(M_y, h_a/h_a^*)$ = -0.98
- $\rho(M_y, k_a/k_a^*)$ = +0.10

Arm damping almost entirely controls the reaching-margin boundary: higher damping dissipates the chirp-induced amplification and causes target shortfall.

Observing the reaching map confirms the same interpretation.
The Spearman coefficient with arm damping is about -0.98, while the correlation with mass and stiffness is much weaker.
Physically, higher damping absorbs the energy of the chirp signal, reducing the motion needed to hit the target, proving that the arm's robustness boundary is almost entirely controlled by damping.

Initial Conditions – Stress-Test Setup

Purpose

- Test the fixed nominal BO command from non-zero initial states
- Keep **mechanical parameters unchanged**
- Keep **optimized arm and chirp unchanged**

Severity Level s_{IC}	Position scale **Ps [m]**	Velocity scale **Vs [m/s]**
1	0.005 A	0.25 $\dot{x}_{ref,0}$
2	0.010 A	0.5 $\dot{x}_{ref,0}$
3	0.025 A	1.00 $\dot{x}_{ref,0}$
4	0.1 A	2.00 $\dot{x}_{ref,0}$
5	0.5 A	3.00 $\dot{x}_{ref,0}$

Initial state variables

- $y_0, x_{b0}, \dot{y}_{b0}, \dot{x}_{b0}$ imposed
- For each severity level:
  - $y_0, x_{b0} \in \{-P_s, 0, P_s\}$
  - $\dot{y}_{b0}, \dot{x}_{b0} \in \{-V_s, 0, V_s\}$

$\longrightarrow$ $5 * 3^4 = 405$

true-EoM simulations

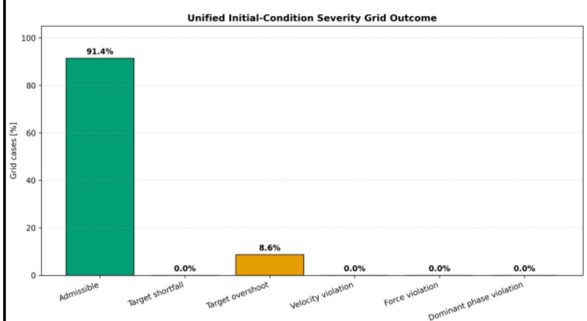
Reference scales

- A = 0.099756 m, nominal BO chirp amplitude
- $\dot{x}_{ref,0} = 2\pi f_{min} A \approx 0.27\ m/s$

The severity index provides a compact way to test **increasingly aggressive initial** displacement and velocity **perturbations**.

For our final test, we keep all mechanical parameters fixed and we start the simulation with different initial conditions, both on position and velocities.
We define a severity index to control how large the perturbations are,
and this results in a total amount of 405 simulations to test all the configurations.

Initial Conditions – Dominant Failure Modes



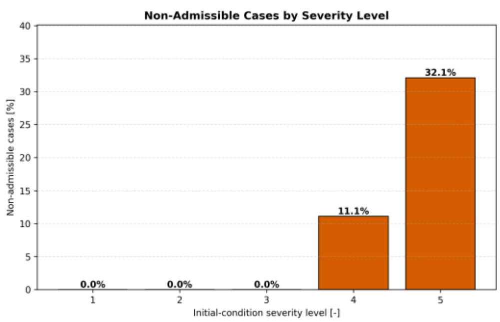
Failure interpretation

- Target overshoot is the only observed failure mode
- No target shortfall occurs
- Velocity and force limits are never activated
- Phase violation is negligible

Initial-condition robustness is not limited by loss of reachability, but by excessive target-band overshoot in the most aggressive initial states.

The test reveals a completely different failure mechanism.
The only failure mode is target overshoot and only for the most agressive initial states.
This means that the command is still low intensity in terms of actuation, but some initial states already favour the same aplification direction of the chirp.

Initial Conditions – Severity Validity Domain



Validity domain

- $s_{IC} \leq 3$ $\longrightarrow$ all tested initial-condition combinations are admissible

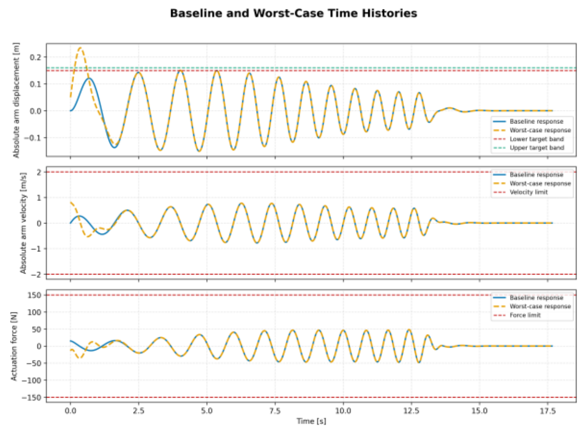
Failure onset

- $3^4 = 81$ cases tested for each severity level
- $s_{IC} = 4 \longrightarrow 9/81$ non admissible cases
- $s_{IC} = 5 \longrightarrow 26/81$ non admissible cases

The fixed nominal command remains valid up to moderate severity; failures appear only for aggressive initial states that already favour chirp-induced amplification

Here we can see the severity map, and our limits are immediately clear.
All tested conditions are admissible up to level 3, and failure only appears at levels 4 and 5.
So we can conclude that the system is robust to moderate states, but extreme initial conditions push it into excessive amplification.

Initial Conditions – Worst Case Analysis



Worst case initial state

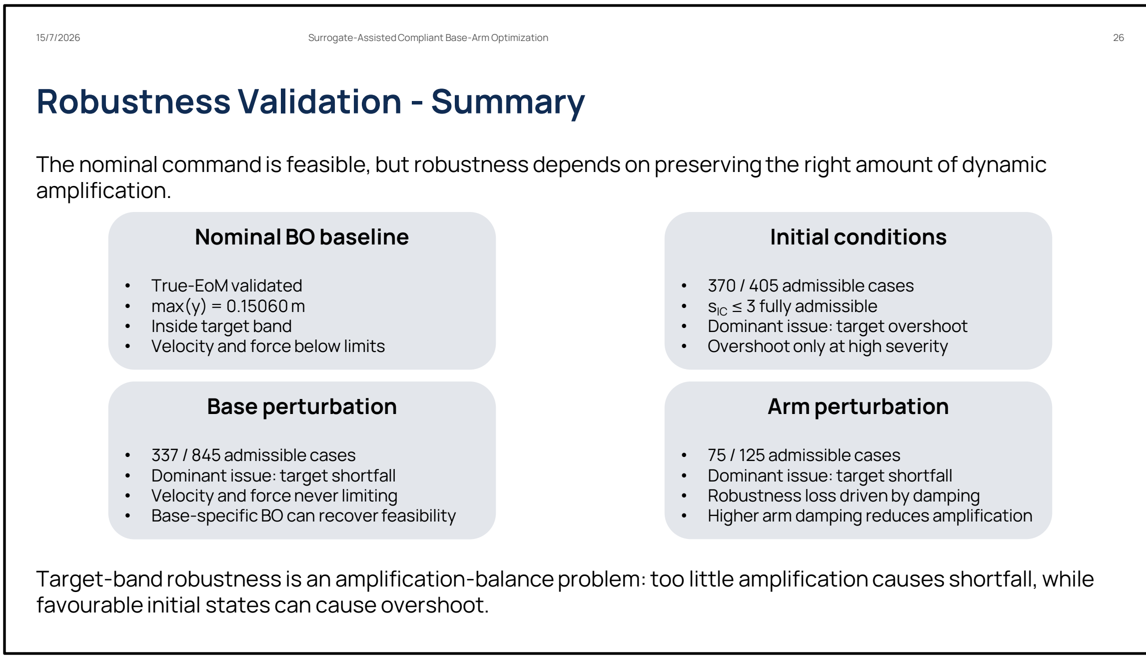
- $s_{IC} = 5$
- $y_0 = +49.88\ \text{mm}$ $\qquad \dot{y}_0 = +0.820\ \text{m/s}$
- $x_{b0} = -49.88\ \text{mm}$ $\qquad \dot{x}_{b0} = +0.820\ \text{m/s}$
- $\max_t(y) = 0.235\ \text{m} \longrightarrow$ **Failure: target overshoot**

Physical explanation

- Large positive relative displacement (large initial force)
- Positive absolute velocities
- Initial state already aligned with chirp amplification

The worst case confirms that initial-condition failures are caused by excessive target-band amplification, while velocity and force remain admissible.

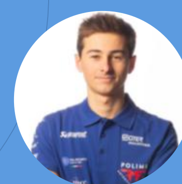
Then we analyse the worst case scenario.
It starts with a initial displacement and velocities that perfectly align with the upcoming chirp signal.
This alignment creates a massive compounding effect.
The resulting motion easily exceeds our upper limit resulting in a excessive amplification.



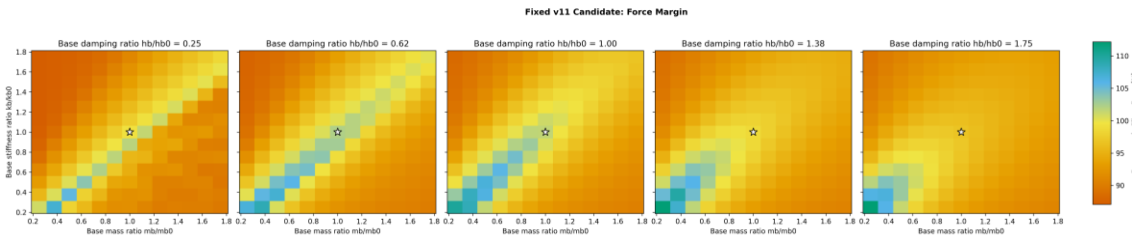
Finally, to summarize, our **nominal** command works, but it **requires** a delicate balance of dynamic amplification
Base perturbations cause – shortfall, that we can fix by a re-optimization.
Arm perturtbations also cause shortfall, and in this case the main limitation is the damping.
Finally extreme initial condition cause overshoot.
An in important thing to say, is that velocity and force limits were never the problem.
So in conclusion to have a robust system the most important parameter to tune is the damping.
With too much we cause a shortfall and with too little we cause a shortfall.

Thanks for your attention!





Base Perturbation- Force Margin



Margin definition

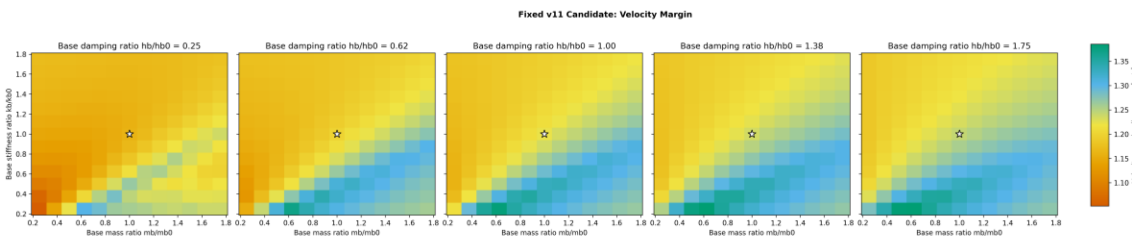
- $M_F = F_{lim} - \max_t(F)$
- $F_{lim} = 150\ N$
- $\boldsymbol{M_F > 0} \longrightarrow$ force below limit
- $\boldsymbol{M_F < 0} \longrightarrow$ force above limit

Spearman correlation

- $\rho(M_F, m_b/m_{b,0}) =$ -0.05
- $\rho(M_F,\ h_b/h_{b,0}) =$ -0.15
- $\rho(M_F, k_b/k_{b,0}) =$ +0.12

Force remains safely below the limit across the full stress-test grid, confirming that force saturation is not the limiting mechanism.

Base Perturbation- Velocity Margin



Margin definition

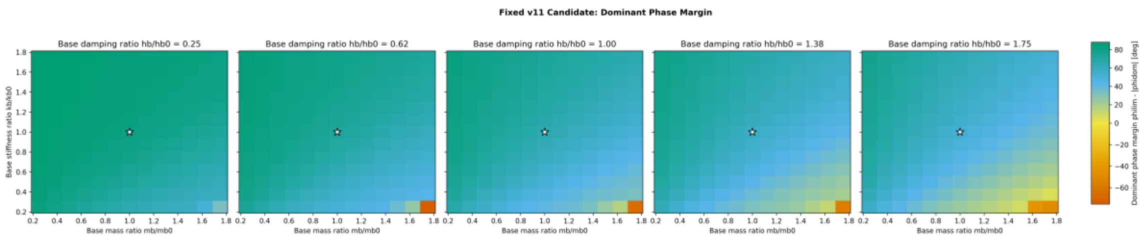
- $M_v = v_{lim} - \max_t(|v|)$
- $v_{lim} = 2\ m/s$
- $\boldsymbol{M_v > 0} \longrightarrow$ **velocity below limit**
- $\boldsymbol{M_v < 0} \longrightarrow$ **velocity above limit**

Spearman correlation

- $\rho(M_F, m_b/m_{b,0}) =$ +0.51
- $\rho(M_F,\ h_b/h_{b,0}) =$ +0.46
- $\rho(M_F, k_b/k_{b,0}) =$ -0.54

Velocity remains safely below the limit across the full stress-test grid; stiffer bases produce more dynamically aggressive responses, but not enough to activate the constraint.

Base Perturbation- Phase Margin



Margin definition

- $M_\phi = \phi_{\text{lim}} - |\Delta\phi_{y,b}(f_{dom})|$
- $\phi_{lim} = 90°$
- Evaluated at the dominant FRF frequency
- $\boldsymbol{M_\phi > 0} \longrightarrow$ **phase below limit**
- $\boldsymbol{M_\phi < 0} \longrightarrow$ **phase above limit**

Spearman correlation

- $\rho(M_F, m_b/m_{b,0}) =$ -0.53
- $\rho(M_F,\ h_b/h_{b,0}) =$ -0.64
- $\rho(M_F, k_b/k_{b,0}) =$ +0.5

Phase is a secondary limitation: violations are rare and occur in already target-shortfall configurations.